

Registration Number : 21BC04132

JECRC UNIVERSITY
I In-Sem Examination October-2023
B.Tech (CSE), Semester-V

Subject: Computer Graphics (BCO 007A)

Time: 75 Min

Maximum marks: 40

Instructions:

1. Attempt all the questions.
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write relevant question numbers before writing the answer.
4. Calculator is allowed.

Course Outcomes:

CO1: Understand the structure of modern computer graphics system

CO2: Understand the basic principles of implementing computer graphics primitives.

SECTION-A

(2*2=4 marks)

1. [CO1] Explain horizontal & vertical retrace in computer graphics display.
2. [CO2] What is the importance of 8 way symmetry in scan conversion of circle.

SECTION-B

(2*7=14 marks)

1. [CO1] Difference between Random Scan and Raster Scan display.
2. [CO2] Write Cohen-Sutherland algorithm for line clipping.

SECTION-C

(2*11=22 marks)

1. [CO1] Explain the working of CRT monitor using suitable diagram.
2. [CO2] Explain DDA Line drawing algorithm. Find the points on line between (10,12) and (16,16) using Bresenham's line drawing algorithm.

Registration Number:

JECRC UNIVERSITY
II In-Sem Examination November-2023
B. Tech (CSE), Semester-V
Subject: Computer Graphics (BCO 007A)

Time: 75 Min.

Maximum marks: 40

Instructions:

1. Attempt all the questions.
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write relevant question numbers before writing the answer.
4. Calculator is allowed.

Course Outcomes:

CO3: Analyze and apply clipping algorithms and transformation on 2D images

CO4: Explore projections and visible surface detection techniques for display of 3D scene on 2D screen

SECTION-A (2*2=4 marks)

1. [CO3] What is "Transformation" in terms of Computer Graphics?
2. [CO4] Write the Homogeneous Matrix for Rotation in X-Direction and Y- Direction for 3D object.

SECTION-B (7*2=14 marks)

1. [CO3] Perform rotation of triangle A(4,1), B(5,2) and C(4,2) by angle 90 degree about the origin.
2. [CO4] Explain the scaling concept in term of 3D object.

SECTION-C (11*2=22 marks)

1. [CO3] Consider a square (0,0), (0,5), (5,0), (5,5). Perform the composite transformations of the square by using the following steps:
 - a. Scale by using $S_x=3$ and $S_y=2$.
 - b. Rotate by angle 45 degree in clockwise direction.
 - c. Translate using $T_x= 4$ and $T_y=5$.
2. [CO4] Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XY plane and find out the new coordinates of the object.

Time: 3hrs

Instructions:

1. Attempt all the questions.
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write relevant section and question numbers before writing the answer.
4. Use of calculator is allowed in this paper.

Course Outcomes:

- CO1: Understand the basics of computer graphics, different graphics systems and applications of computer graphics
CO2: Apply and compare the algorithms for drawing 2D images
CO3: Analyze and apply clipping algorithms and transformation on 2D images
CO4: Explore projections and visible surface detection techniques for display of 3D scene on 2D screen
CO5: Understand visible-surface determination and hidden surface elimination for creating computer graphics applications

Section –A (5*2=10 Marks)

1. [CO1] What is frame buffer? What is horizontal and vertical retrace?
2. [CO2] How world coordinate system is converted to screen coordinate system?
3. [CO3] What is the need of homogeneous coordinates?
4. [CO4] Explain all type of shearing in 3D with transformation matrices.
5. [CO5] Give one advantage and disadvantage of parallel projection.

Section –B (5*7=35 Marks)

1. [CO1] Explain in detail about the Bresenham's Line drawing scan conversion algorithm?
2. [CO2] How flood fill and boundary fill methods work? Explain the Cohen Sutherland algorithm used for Line clipping.
3. [CO3] What are the different transformations in 2-D graphics? Explain each with their matrix representations.
4. [CO4] Translate the given point P(10,10,10) into 3D space with translation factor T(10,20,5).
5. [CO5] Differentiate between Image Space Method and Object Space Method.

Section –C (5*11=55 Marks)

1. [CO1] What is Color CRT Display? Differentiate between Beam-penetration and Shadow-mask methods.
or
[CO1] Write a short note on refresh buffer. Explain the architecture of Vector and Raster Display systems using suitable diagrams.
2. [CO2] Write down and explain the midpoint circle drawing algorithm. Apply it to predict the pixels in any octant for the circle whose center is origin and radius is 8 units.
or
[CO2] Explain DDA algorithm for Line Drawing. Calculate value of pixels which are scan converted by DDA algorithm where endpoints of line are (4, 4) and (11, 8).
3. [CO3] What do you mean by transformation? Given a square with diagonal vertices at (1,1) and (5,5). Perform the following transformations in sequence:
 - a. Scale it to 3 times its size.
 - b. Rotate it by 45-degree.
 - c. Translate it upward by 2 units and 3 units to left
 - d. Reflect it along the x-axis in clock-wise direction
or
[CO3] Explain how region codes are derived in Cohen Sutherland algorithm. Given a clipping window A(20, 20), B(60, 20), C(60, 40), D(20, 40). Using Cohen Sutherland algorithm, find the visible portion of line segment joining the points
 - a. P(30, 80) and Q(100, 30)
 - b. P(10, 30) and Q(80, 30)
 - c. P(70, 10) and Q(10, 50)
4. [CO4] A 3D triangle with vertices A(3,4,1) B(6,4,2), C(5,6,3) is reflected on XZ plane. Find the new coordinates of the triangle.
5. [CO5] Explain the following in detail:
 - a. Scan-Line algorithm
 - b. Z-Buffer algorithm

Registration No. 2180011022

JECRC University

1st In- Semester Examination, OCT. 2023

B.TECH CSE, Semester: V

Subject: Design and Analysis of Algorithms (BCO023A)

Time: 75 mins

Maximum Marks: 40

Instructions:

1. Attempt all the questions
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write relevant question numbers before writing the answers.

Course Outcomes:

CO1: Various methods of calculating complexity

CO2: Finding out the best method for different algorithms

Section A (2*2=4 marks)

Q1. [CO1] Arrange the following growth rates in increasing order

$O(n^{1/4})$, $O(n^{1.5})$, $O(n^3 \lg n)$, $O(n^{1.02})$, $\Omega(n^6)$, $\Omega(n!)$, $O(\sqrt{n})$, $O(n^{6/2})$, $\Omega(2^n)$.

Q2. [CO2] Construct a huffman tree by using given nodes. Also mention bits used before and after the construction of huffman tree.

Value	A	B	C	D	E	F
Frequency	5	25	7	15	4	12

Section B (2*7=14 marks)

Q3.[CO1] Consider a list of some elements from which maximum and minimum can be found

1	2	3	4	5	6	7	8	9
50	40	-5	-9	45	90	65	25	75

Q4.[CO2] Find out the LCS of {A = K, A, N, D, L, A, P} and B = {A, N, D, L}

Section C (2*11=22 marks)

Q5.[CO1] Solve recurrence :-

$T(n) = T(n-1) + 1$, here $T(1) = \theta(1)$ solve using iterative method.

$T(n) = T(n-1) + n$, here $T(0) = 0$ solve using backward substitution method.

$T(n) = T(2n/3) + 1$ solve using master method.

Compute Matrix Chain Order.

Q6. [CO2]

Matrix	Dimension
A1	5x4
A2	4x6
A3	6x2
A4	2x7

Registration No:.....

JECRC UNIVERSITY, JAIPUR
2nd In- Semester Examination, 2023
B.TECH CSE, Semester: V
Subject: Design and Analysis of Algorithms
Subject Code: BCO 023A

Time: 75 mins

Maximum Marks: 40

Instructions:

1. Attempt all the questions
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write relevant question numbers before writing the answers.

Course Outcomes:

CO3: Knowledge of Graph and String matching algorithm

CO4: Computational geometry, like Lower bound theory, modular arithmetic and CRT

Section A (2*2=4 marks)

- Q1. [CO3] What is residual capacity in a flow network ?
- Q2. [CO4] Provide the solution for 4-queens problem.

Section B (2*7=14 marks)

Q3.[CO3] Explain Bad Character Heuristics in Boyer Moore Algorithm for text $T = \text{nyoo}$ and pattern $P = \text{noyo}$ $n = 8$

Q4.[CO4] State Chinese Remainder Theorem. Solve the following congruence.

$$X \equiv 2 \pmod{3}$$

$$X \equiv 3 \pmod{5}$$

$$X \equiv 2 \pmod{7}$$

Section C (2*11=22 marks)

Q5.[CO3] Using KMP Matching Algorithm find whether pattern $P = \text{aabab}$ is present in the text $T = \text{aaababaabaa}$

Q6. [CO4] a) Find the minimum set cover and minimum cost of set cover.

$$U = \{1, 2, 3, 4, 5\}$$

$$S = \{S_1, S_2, S_3\}$$

$$S_1 = \{4, 1, 3\}, \text{Cost}(S_1) = 5$$

$$S_2 = \{2, 5\}, \text{Cost}(S_2) = 10$$

$$S_3 = \{1, 4, 3, 2\}, \text{Cost}(S_3) = 3$$

Why is minimum set cover useful?

b) Differentiate between Las Vegas and Monte Carlo Algorithms through example. [5+6=11 Marks]

n	o	y	*
3	4	1	4

$$v_1 = 4 - 2 - 1 = 1$$
$$v_n = 4 - 0 - 1 = 3$$

$n = 3$

Time: 3 hrs

Instructions:

1. Attempt all the questions.
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write relevant question numbers before writing the answer.

Course Outcome (CO):

- CO1: Various methods of calculating complexity
CO2: Finding out the best method for different algorithms
CO3: Knowledge of Graph and String matching algorithm
CO4: Computational geometry, like Lower bound theory, modular arithmetic and CRT
CO5: Various Decision Problems like NP Complete, NP hard

SECTION-A

[2*5=10 mark each]

Q 1- Answer the following questions.

- I) (CO1)(co1) What will be time complexity of below code :

```
for I = 1; I * I < N; i++
    for J = N; J > 1; J=J/K
        printf " JECRC";
```
- II) (CO2) How many bits may be required for encoding the message 'Mississippi'
- III) (CO3) When a forward or backward edge may includes in augmented path
- IV) (CO4) Why prime no is prefers to choose as key in Cryptography ?
- V) (CO5) State the set cover problem.

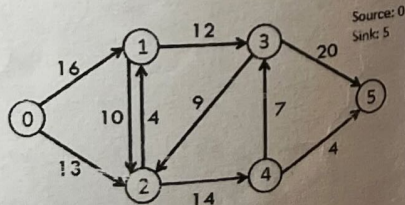
SECTION-B

[7*5=35 mark each]

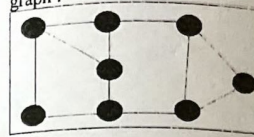
Q 2- Answer the following questions.

- I) (CO1) Write the down max-min algorithm and apply to find maximum and minimum number for following list using Divide & Conquer approach :
50,43,77,8,99,56,73,25,49,97
Also draw proper recursive tree .
- II) (CO2) Differentiate the Greedy Approach and Dynamic Programming Approach to solve a problem. Solve below problem using 0/1 knapsack approach: $N=3$; $w=(5,4,3)$, $v=(6,5,4)$ and $W=7$; [2+5]
- III) (CO3) Explain following terms : a) Residual Capacity b) Augmented path [2+5]

Apply Ford Fulkerson Algorithm to find out maximum flow from Source to Sink -



- IV) (CO4) what is vertex cover problem? find the vertex cover of the following graph :



- V) (CO5) Explain P, NP, NPH and NPC class of problems with an example. Also draw the relationship Diagram.

SECTION-D

Q 4- Answer the following questions.

[11*5=55 mark each]

- I) CO1 Find out the asymptotic bound of below recurrence :
 $[2.5+2.5+3+3]$
 (i) $T(n) = 4T(n/4) + O(n^2)$ (Master Theorem)
 (ii) $T(n) = 2T(n/2) + n / \log n$ (Master Theorem)
 (iii) $T(n) = 2T(n^{1/2}) + \log n$ (Substitution)
 (iv) $T(n) = T(n-1) + 1$ (Iteration Method)
- II) CO2 Write down the dynamic programming. Also apply this approach on below given problem : Given a chain of four matrices A_1, A_2, A_3 and A_4 , with $P_0=5, P_1=4, P_2=6, P_3=2$, and $P_4=7$. [3+8]
- III) CO3 Text = DDBDDBADECB and Pattern = ECB, for above text and pattern, implement Rabin Karp Algorithm. Also mention all the spurious hit, and define Rolling Hash Function. (7+2+2)
- IV) (CO4)(a) Differentiate Las Vegas and Monte Carlo Randomized approach with an example? [6+5]
 (CO4)(b) Explain by an example, how the Las Vegas Approach improves the Complexity of Quick Sort.
- V) (CO5) Explain the concept of Cook's Theorem using suitable example. Explain when an NP hard problem can be considered in category of NPC? [8+3]

Registration No: 21BP0N132

JECRC UNIVERSITY, JAIPUR

I In- Semester Examination October 2023

B. Tech. V Semester Computer Science Engg.

Subject: Formal Languages & Automation Theory

Time: 1.15 hr.

Subject Code: BCO 017A

Maximum Marks: 40

CO1: Understand and construct finite state machines and the equivalent regular expressions.

CO2: Prove the equivalence of languages described by FSM and regular expressions.

Section -A (Very Short answers)

(2*2=4 Marks)

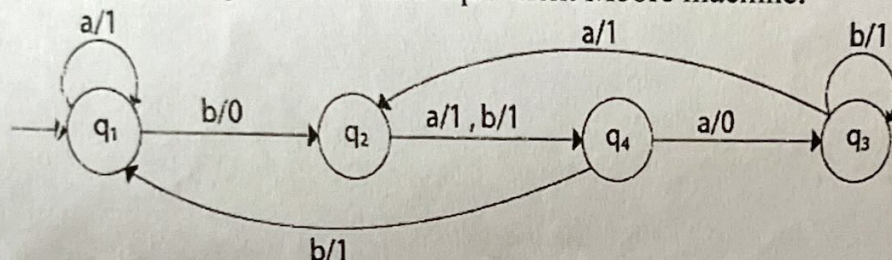
CO1)1: Construct DFA for all set of strings which starts and ends with different symbol where $\Sigma = (0,1)$.

CO2)2: What is the regular expression for $\Sigma = (0,1)$ where $|W| \equiv 2 \pmod{3}$.

Section - B (Short answers)

(7*2=14 Marks)

CO1)1: Convert the following Mealy machine into equivalent Moore machine.



CO2)2: Find the regular expression and transition diagram for the following languages $\Sigma = (a,b)$:

a) Each Ending having last two symbols same.

b) 19th symbol is "a" while reading the string from RHS.

Section -C (Long Answers)

(11*2=22 Marks)

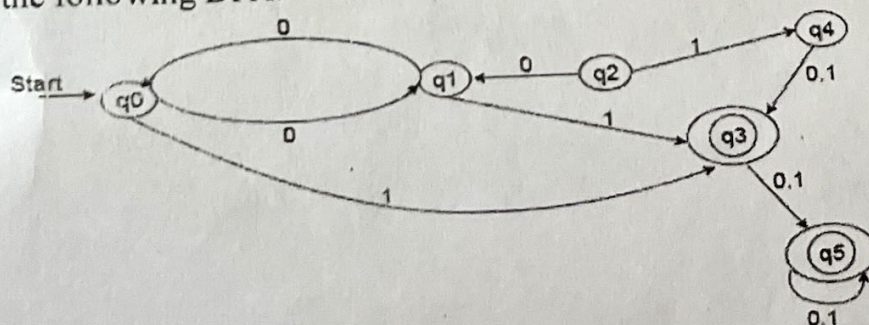
CO2)1(A): Construct Finite Automata for the $(a+b)^*(ba+aa)(a+b)^*$

1 (B) What is Arden's theorem? And prove the following-

$$(1*1+1*100^*) + (10*1+0*10)(1+00^*)(10*1+0*10)^* = 1*10^*(10*1+0*10)$$

CO1) 2 (A): Differentiate DFA & NDFA.

2 (B): Minimize the following DFA:



Registration No:

JECRC UNIVERSITY, JAIPUR
II In-Semester Examination October 2023
B. Tech V Semester
Subject: Formal Languages & Automation Theory
Subject Code: BCO 017A

Time: 1.15 hr.

Maximum Marks: 40

CO3: Construct pushdown automata and the equivalent context free grammars.

CO4: Prove the equivalence of languages described by pushdown automata and context free grammars.

Section -A (Very Short answers)**(2*2=4 Marks)**Q1[CO3] Q3. G is $S \rightarrow aS|bS|a|b$ Find $L(G)$

Q2[CO4] What are the two approaches of language acceptance in PDA?

Section - B (Short answers)**(7*2=14 Marks)**

Q3[CO3] Write technical definition of leftmost derivation and rightmost derivation. Solve the below given example for leftmost derivation and rightmost derivation and draw trees for both derivations respectively.

 $S \rightarrow 0B|1A$ $A \rightarrow 0|0S|1AA$ $B \rightarrow 1|1S|0BB$ String is $w = 00110101$

Q4 [CO4] Explain block diagram of push down automata. And Differentiate with Finite Automata.

Section -C (Long Answers)**(11*2=22 Marks)**

Q5[CO3] a) Find a grammar in Chomsky normal form equivalent to-

 $S \rightarrow aAbB, A \rightarrow aA|a, B \rightarrow bB|b.$ b) L be the set of all Palindromes over $\{0, 1\}$. Construct Grammar.

Q6[CO4] Explain PDA and its operation. Construct the PDA for the following languages.

 $\{a^n b^{n+m} c^m \mid n \geq 1, m \geq 1\}$

BCON 132

End-Semester Examination Paper
 School of Engineering
 End-Semester Examination 2023-2024
 B.Tech (CSE), Semester-V
 Subject: FLAT
 Subject Code: BCO 017A
 Maximum Marks: 100

22/12
 11

Time: 3 hrs

Course Outcome (CO):

- At the ends of this course students will have:
- CO1: Understand and construct finite state machines and the equivalent regular expressions.
 - CO2: Prove the equivalence of languages described by pushdown automata and context free grammars.
 - CO3: Construct pushdown automata and the equivalent context free grammars.
 - CO4: Prove the equivalence of languages described by pushdown automata and context free grammars.
 - CO5: Construct Turing machines and Post machines and prove the equivalence of languages described by Turing machine and Post machines.

1. Attempt all the questions.
2. Illustrate your answers with suitable examples and diagrams, wherever necessary.
3. Write relevant question numbers before writing the answer.

Section -A (Very Short answers)

(2*5=10 Marks)

1. [CO1] Consider the finite state machine whose transition function δ is given in below table. Here, $Q = \{q_0, q_1, q_2, q_3, \Sigma = \{0, 1\}, F = \{q_0\}$. Give the entire sequence of states for the input string 110101.

State/ Σ	Input=0	Input=1
$\rightarrow q_0$	q_2	q_1
q_1	q_3	q_0
q_2	q_0	q_1
q_3	q_1	q_2

2. [CO2] Let $G = (\{S, A_1, A_2\}, \{a, b\}, P, S)$ where P consists of $S \rightarrow aA_1A_2b, A_1 \rightarrow baA_1A_2b, A_2 \rightarrow A_1ab, aA_1 \rightarrow baa, bA_2 \rightarrow abab$. Test if $w = baabbabaabbaba$ is in $L(G)$.
3. [CO3] What is derivation tree? Give example.
4. [CO4] Can we process the string of the form $a^n b^n$ by the simple finite automaton? If yes then how? If no then why?
5. [CO5] What is Turing Machine? Explain its tuples.

(7*5=35 Marks)

Section - B (Short answers)

1. [CO1] What is mealy machine and moore machine. Convert moore to mealy machine

Present state	Next State		Output
	a=0	a=1	
$\rightarrow q_0$	q_1	q_2	1
q_1	q_1	q_2	0
q_2	q_2	q_1	1
q_3	q_0	q_1	1

2. [CO2] (a) Explain closure properties of language classes
 (b) Describe the following sets by regular expressions:
 (b 1) L_1 = the set of all strings of 0's and 1's ending in 00.
 (b 2) L_2 = the set of all strings of 0's and 1's beginning with 0 and ending with 1.
 (b 3) $L_3 = \{A, 11, 1111, 111111, \dots\}$

3. [CO3] Write technical definition of leftmost derivation and rightmost derivation. Solve the below given example for leftmost derivation and rightmost derivation and draw trees for both derivations respectively
 $S \rightarrow 0B11A$
 $A \rightarrow 00S11AA$
 $B \rightarrow 11S0BB$
 String is $w = 00110101$

4. [CO4] Explain block diagram of push down automata
5. [CO5] Consider the Turing machine Description given in table. Draw the computation sequence of the input string 00

State	Tape symbol		
	b	0	1
$\rightarrow q_1$	1L q_2	0R q_1	
q_2	bR q_1	0L q_2	1L q_2
q_3		bR q_2	bR q_3
q_4	0R q_3	0R q_4	1R q_4
q_5	0L q_3		

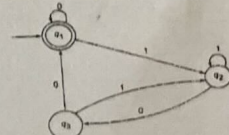
Section -C (Long Answers)

(11*5=55 Marks)

1. [CO1] Construct the minimum state automaton equivalent to a DFA whose transition table is given below:

State	a	b
$\rightarrow q_0$	q_1	q_2
q_1	q_4	q_3
q_2	q_4	q_3
q_3	q_5	q_6
q_4	q_7	q_6
q_5	q_3	q_6
q_6	q_6	q_6
q_7	q_4	q_6

2. [CO2] (a) State and prove Arden's Theorem and find out Regular Expression for the following FA



- b) Construct Finite Automata for the $ab + (ba+ab)^*a$
3. [CO3] a) Prove that two regular expressions are equivalent. $(a+b)^* = a^*(ba^*)^*$
 b) Let $G = (V, \Sigma, P, S)$ be the given productions $S \rightarrow AB, A \rightarrow a, B \rightarrow b, C \rightarrow c$. Find G^* such that every variable in G^* derive some terminal string
4. [CO4] Construct a pda (push down automata) accepting all palindromes over $\{a, b\}$
5. [CO5] Explain block diagram of Turing machine. Discuss following questions as well with the help of example
 a) Represent Turing machine through ID (instantaneous description)
 b) Represent Turing machine by transition table
 c) Represent Turing machine by transition system